

Model Error Resonance: The Geometric Nature of Error Dynamics[★]

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Abstract

This paper introduces a geometric theory of model error, treating true and model dynamics as geodesic flows generated by distinct affine connections on a smooth manifold. When these connections differ, the resulting trajectory discrepancy—termed the Latent Error Dynamic Response (LEDR)—acquires an intrinsic dynamical structure governed by curvature. We show that the LEDR satisfies a Jacobi-type equation, where curvature mismatch acts as an explicit forcing term. In the important case of a flat model connection, the LEDR reduces to a classical Jacobi field on the true manifold, causing Model Error Resonance (MER) to emerge under positive sectional curvature. The theory is extended to a discrete-time analogue, establishing that this geometric structure and its resonant behavior persist in sampled systems. A closed-form analysis of a sphere–plane example demonstrates that curvature can be inferred directly from the LEDR evolution. This framework provides a unified geometric interpretation of structured error dynamics and offers foundational tools for curvature-informed model validation.

Key words: Affine connections; curvature; geodesic deviation; Jacobi fields; model error resonance; discrete-time systems; geometric dynamics.

1 Introduction

Model error is typically regarded as a static discrepancy between the predicted and true outputs of a dynamical system. This viewpoint is prevalent in system identification, model-based control, state estimation, and learning-based prediction, where residuals or one-step-ahead errors are used to assess model quality.

Yet, this static perspective is often insufficient. In complex engineering scenarios, such as high-performance flight dynamics or coupled non-linear systems, it is frequently observed that models with low static residuals can still exhibit distinct, structured oscillatory error responses. These oscillations are rarely random noise; rather, they serve as a spectral signature of the underlying structural mismatch. We interpret this phenomenon as a form of ‘geometric friction’: a dynamical tension that arises when a model attempts to track a physical system evolving on a topologically distinct manifold. The persistence of these structured errors suggests that the discrepancy is not merely numerical, but intrinsic to the curvature difference between the model and reality.

However, when both the model and the physical system evolve as flows generated by distinct affine connections

on a manifold, model error necessarily acquires a *dynamical* structure governed by curvature. This observation motivates a geometric reinterpretation of model fidelity that [5,10,13,15].

In this work, we study two dynamical systems defined on the same smooth manifold but endowed with different affine connections. The true system evolves along geodesics of a connection ∇^t , while the model system evolves along geodesics of another connection ∇^m [9,7]. Although the two systems share identical initial conditions, the resulting trajectories generally separate due to differences in the curvature and connection structure. We refer to the trajectory discrepancy as the *latent error* and denote it by $\xi(t)$. Rather than treating $\xi(t)$ as a static residual, we analyze its evolution using the geometric machinery of geodesic deviation and Jacobi fields [5,14,2].

Our main result shows that the latent error satisfies a Jacobi-type equation. In continuous time, the acceleration of $\xi(t)$ is governed by the curvature tensor of the true connection, together with correction terms arising from the difference between ∇^t and ∇^m . When the model connection is flat, the latent error reduces to a Jacobi field on the true manifold, and model error resonance emerges automatically under positive sectional curvature [13,14]. This provides a geometric explanation for persistent structured error dynamics commonly observed in model-based predictors and sampled-data im-

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plementations.

We further develop a discrete-time analogue of the theory. By discretizing the covariant derivatives along the model trajectory, we obtain a second-order difference equation that preserves the structure of the continuous Jacobi operator, in line with ideas from discrete geometric mechanics [12]. The resulting discrete latent error dynamics exhibit the same curvature-driven effects as in continuous time, including resonance under positive sectional curvature. Thus, the geometric interpretation extends naturally to sampled systems.

To illustrate the theory, we analyze in closed form the case where the true system evolves on a sphere of constant curvature while the model assumes a flat Euclidean plane. The LEDR dynamics reduce to a harmonic oscillator with frequency determined by the curvature, yielding an analytically tractable and physically transparent example of curvature-induced resonance [5,10]. This example demonstrates that the proposed framework can reveal curvature properties directly from the latent error evolution, even in the absence of explicit knowledge of the true connection.

Overall, this work establishes a geometric foundation for understanding model error as a dynamical quantity governed by curvature and geodesic deviation. The resulting framework connects differential geometry, geometric mechanics, and dynamical-systems stability theory [1,4,2,11] and provides a basis for further development in model validation, consistency analysis, and curvature-informed dynamical modeling.

The paper is organized as follows. Section 2 introduces geometric preliminaries and formulates the problem. Section 3 derives the continuous LEDR equation. Section 4 presents the discrete analogue. Section 5 characterizes curvature-induced resonance. Section 6 provides a closed-form constant-curvature example. Additional derivations and technical results are given in the appendices.

2 Geometric Framework and Problem Formulation

This section establishes the geometric framework for our analysis. We introduce the necessary concepts from differential geometry, formally define the true and model dynamical systems as geodesic flows governed by distinct affine connections, define the latent error as a geodesic deviation field, and state the central problem addressed in this paper. Our notation follows standard references [9,7,16].

2.1 Manifolds, Affine Connections, and Geodesics

Let $\mathcal{M}$ be a smooth n -dimensional manifold representing the state space. For a smooth curve $\gamma(t) \in \mathcal{M}$ with velocity vector $T(t) = \dot{\gamma}(t)$, the covariant derivative of a vector field $V(t)$ along $\gamma(t)$ is denoted by $\frac{DV}{dt} := \nabla_{T(t)} V$, where ∇ is a torsion-free affine connection [9,5]. In local coordinates $\{x^i\}$, ∇ is specified by its Christoffel symbols Γ_{jk}^i such that $\nabla_{\partial_j} \partial_k = \Gamma_{jk}^i \partial_i$.

A curve $\gamma(t)$ is a geodesic of the connection ∇ if it parallel transports its own tangent vector, i.e.,

$$\frac{DT}{dt} = \nabla_T T = 0.$$

Geodesics represent the straightest possible paths on the manifold and correspond to the free-motion trajectories of a dynamical system whose geometry is described by ∇ [10,1].

2.2 True and Model Geodesic Flows

We consider two dynamical systems evolving on $\mathcal{M}$ but governed by different geometric structures.

- The **true system** generates a trajectory $\gamma_t(t)$, which is a geodesic of a "true" affine connection ∇^t with Christoffel symbols $\Gamma_{jk}^t(x)$.

$$\nabla_{T_t}^t T_t = 0, \quad \text{where } T_t = \dot{\gamma}_t.$$

- The **model system** generates a trajectory $\gamma_m(t)$, which is a geodesic of a "model" connection ∇^m with symbols $\Gamma_{jk}^m(x)$. We idealize the model's predictive mechanism as a **shadow integrator** that produces this geodesic flow.

$$\nabla_{T_m}^m T_m = 0, \quad \text{where } T_m = \dot{\gamma}_m.$$

Both systems are assumed to start from identical initial conditions: $\gamma_t(0) = \gamma_m(0)$ and $T_t(0) = T_m(0)$. This formulation is consistent with geometric mechanics and control theory [4,1].

2.3 Latent Error as a Geodesic Deviation Field

The discrepancy between the two trajectories gives rise to the central object of our study: the **Latent Error Dynamic Response (LEDR)**. Intuitively, in a local coordinate chart, the LEDR displacement vector is simply

$$\xi(t) = x_t(t) - x_m(t).$$

More formally and intrinsically, the LEDR is a vector field $\xi(t)$ defined along the model trajectory $\gamma_m(t)$. It represents the vector pointing from $\gamma_m(t)$ to $\gamma_t(t)$. This

can be rigorously defined using the exponential map and parallel transport:

$$\xi(t) = \text{PT}_{\gamma_t(t) \rightarrow \gamma_m(t)} \left(\exp_{\gamma_m(t)}^{-1}(\gamma_t(t)) \right) \in T_{\gamma_m(t)} \mathcal{M},$$

where PT and exp are associated with the true connection ∇^t . This definition ensures that $\xi(t)$ is a tangent vector at the model state, representing the *latent dynamic inconsistency* between the two systems. From a geometric perspective, $\xi(t)$ is a **geodesic deviation field** between two flows generated by distinct connections, analogous to classical Jacobi fields which measure the deviation between geodesics from the same connection [5,13,14,6].

2.4 Curvature and Connection Mismatch

The geometric discrepancy between the true system and the model is encoded in the difference between their connections and the resulting curvatures. The **connection mismatch** is the tensor field

$$\Delta \Gamma_{jk}^i := \Gamma_{jk}^i(x) - \Gamma_{jk}^m(x).$$

The Riemann curvature tensor of a connection, $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$ [10], measures the non-commutativity of covariant derivatives and thus the intrinsic curvature of the manifold. The difference in curvature tensors,

$$\Delta R = R^t - R^m,$$

plays a central role in the emergence of oscillatory error dynamics. These difference objects are central to affine differential geometry [7,8,16].

2.5 Objective: The Dynamics of Latent Error

The central question of this paper is:

How does the geometric discrepancy encoded in $\nabla^t - \nabla^m$ determine the dynamical evolution of the LEDR field $\xi(t)$?

Our analysis will show that $\xi(t)$ satisfies a perturbed Jacobi equation, where the curvature of the true manifold acts as a restoring force and the connection mismatch acts as a forcing term. This framework reveals that phenomena like Model Error Resonance (MER) are not numerical artifacts but are instead fundamental consequences of the underlying geometry.

3 Main Theory

This section presents the main theoretical results of the paper. We show that the LEDR vector field satisfies a perturbed Jacobi equation, establish structural

conditions under which Model Error Resonance (MER) necessarily appears, and derive geometric lower bounds demonstrating that persistent curvature mismatch cannot be compensated by any model dynamics. These results collectively establish LEDR as a dynamically measurable indicator of model fidelity.

3.1 Preliminaries on covariant derivatives

Let $\gamma_m(t)$ be the model trajectory and $T^m = \dot{\gamma}_m(t)$ its tangent vector. The covariant derivative along γ_m induced by the true connection ∇^t is

$$\frac{D\xi}{dt} := \nabla_{T^m}^t \xi, \quad \frac{D^2\xi}{dt^2} := \nabla_{T^m}^t \nabla_{T^m}^t \xi.$$

The curvature operator is defined as [10,5]

$$R^t(X, Y)Z := \nabla_X^t \nabla_Y^t Z - \nabla_Y^t \nabla_X^t Z - \nabla_{[X, Y]}^t Z.$$

The LEDR vector field $\xi(t) \in T_{\gamma_m(t)} \mathcal{M}$ is defined in Section 2.

3.2 LEDR satisfies a perturbed Jacobi equation

The first theorem shows that LEDR satisfies a geodesic deviation equation with a forcing term induced by the connection mismatch.

Theorem 1 (LEDR–Jacobi Equivalence). *Let $\gamma_t(t)$ and $\gamma_m(t)$ denote the true and model trajectories. Assume $\|\xi(t)\|$ is small and both trajectories lie in a common coordinate chart. Following standard linearization arguments in geodesic variation [5,10], the LEDR vector field satisfies*

$$\frac{D^2\xi}{dt^2} + R^t(T^m, \xi) T^m = F_{\Delta\Gamma}(T^m, \xi) + \mathcal{O}(\|\xi\|^2) + \mathcal{O}(\|\xi\| \|\Delta\Gamma\|), \quad (1)$$

where $\Delta\Gamma = \Gamma^t - \Gamma^m$ and

$$F_{\Delta\Gamma}^i = -\Delta\Gamma_{jk}^i T^m{}^j T^m{}^k.$$

If $\Delta\Gamma$ and higher-order terms are negligible, then ξ satisfies the true Jacobi equation.

Proof. See Appendix A for the full coordinate derivation. $\square$

3.3 MER as curvature-induced resonance

The next result formalizes the condition under which LEDR exhibits oscillatory behavior, establishing MER as a geometric necessity.

Theorem 2 (Curvature-Induced Resonance). *Assume: (i) the model connection is flat $\Gamma^m \equiv 0$; (ii) the true curvature tensor has sectional curvature $K(t) = K(T^m(t), \xi(t))$ along the model trajectory. Then the LEDR dynamics reduce to*

$$\ddot{\xi} + K(t)\xi = 0.$$

If $K(t) \equiv K > 0$ is constant, LEDR exhibits harmonic oscillation

$$\xi(t) = A \sin(\sqrt{K}t) + B \cos(\sqrt{K}t),$$

with natural frequency $\omega = \sqrt{K}$.

This is the classical stability property of Jacobi fields under positive sectional curvature [13,14].

Proof. Flatness of ∇^m implies $\Delta\Gamma = \Gamma^t$ and $R^m = 0$. Substituting into (1) yields the Jacobi equation. Constant curvature yields the scalar form. $\square$

This theorem explains MER as a curvature-restoring phenomenon: the true manifold “pulls” the model trajectory back toward the true geodesic with curvature-dependent stiffness.

3.4 Resonance or divergence is unavoidable under curvature mismatch

We now show that curvature mismatch prevents exponential decay of LEDR. Thus, if the model curvature does not match the true curvature, $\xi(t)$ cannot be uniformly driven to zero.

Theorem 3 (Curvature Obstruction to Error Decay). *Let $K_t(t)$ and $K_m(t)$ denote the sectional curvatures of the true and model connections along $\gamma_m(t)$. If*

$$|K_t(t) - K_m(t)| \geq \kappa_0 > 0 \quad \text{for all } t \in [0, T],$$

then LEDR satisfies the lower bound

$$\|\xi(t)\| \geq C \int_0^t |K_t(s) - K_m(s)| ds - \varepsilon(t),$$

where $C > 0$ depends on the local geometry and $\varepsilon(t)$ collects higher-order terms. In particular, $\|\xi(t)\|$ cannot converge to zero uniformly.

Proof. Subtract the Jacobi operators associated with the two curvatures and apply Gronwall-type inequalities on the resulting second-order differential inequality. See Appendix C. $\square$

This means: **If the true curvature differs from the model curvature, LEDR cannot vanish.** MER is therefore not a numerical artifact—it signals structural mismatch.

3.5 General deviation under arbitrary connection mismatch

Finally, we characterize the LEDR dynamics when neither the true nor the model connection is flat.

Theorem 4 (General LEDR Deviation Law). *Let ∇^t and ∇^m be arbitrary torsion-free connections. Define the curvature mismatch operator*

$$\Delta R := R^t - R^m.$$

Then the LEDR vector satisfies

$$\frac{D^2\xi}{dt^2} + R^m(T^m, \xi)T^m = -\Delta R(T^m, \xi)T^m + F_{\Delta\Gamma}(T^m, \xi) + \mathcal{O}(\|\xi\|^2).$$

Proof. Rewrite (1) by adding and subtracting the model curvature term $R^m(T^m, \xi)T^m$. $\square$

This theorem shows that LEDR encodes both curvature mismatch and connection mismatch, meaning it contains full structural information about the difference between the true and model dynamics.

3.6 Summary of the main theory

The results in this section establish that:

- LEDR is governed by a Jacobi-type deviation equation;
- MER arises from positive sectional curvature in the true system;
- curvature mismatch imposes a fundamental lower bound on model error;
- LEDR encodes complete geometric information about the difference between the true and model dynamics.

These properties justify LEDR as a robust, physically interpretable tool for model-fidelity assessment and dynamical-system explainability.

4 Discrete-Time LEDR

Real-world systems—including flight-data recordings, digital controllers, and neural-network based predictors—operate in discrete time. This section derives the discrete-time analogue of the LEDR dynamics and shows that the resulting error evolution satisfies a second-order difference equation that discretizes the

Jacobi operator. We additionally characterize the numerical stability region and demonstrate that curvature mismatch yields unavoidable persistent oscillation or divergence in the discrete setting.

4.1 Discrete covariant derivatives

Let $\{x_{m,k}\}$ and $\{x_{t,k}\}$ denote the model and true trajectories sampled at time steps $t_k = kh$. Define the discrete LEDR vector

$$\xi_k := x_{t,k} - x_{m,k}.$$

We use the central-difference approximation of the covariant derivative:

$$\left. \frac{D\xi}{dt} \right|_{t_k} \approx \frac{1}{2h} (\xi_{k+1} - \xi_{k-1}),$$

and the discrete second covariant derivative:

$$\left. \frac{D^2\xi}{dt^2} \right|_{t_k} \approx \frac{1}{h^2} (\xi_{k+1} - 2\xi_k + \xi_{k-1}).$$

These expressions are consistent up to $\mathcal{O}(h^2)$ and match standard discrete geometric mechanics constructions.

4.2 Discrete Jacobi operator

Let $T_{m,k}$ be the model discrete velocity

$$T_{m,k} = \frac{x_{m,k+1} - x_{m,k-1}}{2h}.$$

The continuous Jacobi term

$$R^t(T^m, \xi)T^m$$

is discretized via the curvature tensor at $x_{m,k}$:

$$[R_k^t(T_{m,k}, \xi_k)T_{m,k}]^i = R^{ti}{}_{j\ell m}(x_{m,k})T_{m,k}^j \xi_k^\ell T_{m,k}^m.$$

4.3 Discrete LEDR evolution equation

Discretizing the continuous LEDR equation

$$\frac{D^2\xi}{dt^2} + R^t(T^m, \xi)T^m = F_{\Delta\Gamma}(T^m, \xi)$$

yields the second-order difference equation:

$$\begin{aligned} \xi_{k+1} = & 2\xi_k - \xi_{k-1} - h^2 R_k^t(T_{m,k}, \xi_k)T_{m,k} \\ & + h^2 F_{\Delta\Gamma,k}(T_{m,k}, \xi_k) + \mathcal{O}(h^3, \|\xi_k\|^2). \end{aligned} \quad (2)$$

This is the discrete Jacobi operator plus a forcing term due to the connection difference.

4.4 Flat-model simplification

If the model connection is flat ($\Gamma^m \equiv 0$), then $\Delta\Gamma = \Gamma^t$ and

$$F_{\Delta\Gamma,k} = 0 \text{ to first order.}$$

Equation (2) reduces to the discrete Jacobi equation:

$$\xi_{k+1} = 2\xi_k - \xi_{k-1} - h^2 R_k^t(T_{m,k}, \xi_k)T_{m,k}. \quad (3)$$

4.5 Constant curvature case

For constant sectional curvature K , the curvature term satisfies

$$R^t(T, \xi)T = K\xi,$$

yielding the scalar recurrence

$$\xi_{k+1} = 2\xi_k - \xi_{k-1} - h^2 K \xi_k. \quad (4)$$

Define $\lambda = h^2 K$. The recurrence becomes

$$\xi_{k+1} - (2 - \lambda)\xi_k + \xi_{k-1} = 0.$$

Characteristic equation:

$$\mu^2 - (2 - \lambda)\mu + 1 = 0.$$

The roots satisfy

$$\mu_{\pm} = 1 - \frac{\lambda}{2} \pm \sqrt{\frac{\lambda^2}{4} - \lambda}.$$

4.6 Discrete stability region

MER occurs when the dynamics are oscillatory, i.e. when $|\mu_{\pm}| = 1$ or $\mu_{\pm}$ are complex conjugates. This occurs when

$$0 < \lambda < 4 \quad \Longleftrightarrow \quad 0 < K < \frac{4}{h^2}.$$

Thus:

For sufficiently small h , any positive true curvature $K > 0$ produces oscillatory LEDR dynamics (discrete MER).

Furthermore:

- $K > 0 \rightarrow$ oscillation with frequency

$$\omega_d = \frac{1}{h} \arccos\left(1 - \frac{h^2 K}{2}\right)$$

- $K < 0 \rightarrow$ exponential divergence - $K = 0 \rightarrow$ linear drift (flat case)

4.7 Lower bound under discrete curvature mismatch

Let $K_{t,k}$ and $K_{m,k}$ be the discrete curvatures of the true and model connections. If

$$|K_{t,k} - K_{m,k}| \geq \kappa_0 > 0 \quad \forall k,$$

then the discrete LEDR satisfies the inequality

$$\|\xi_k\| \geq Ch \sum_{j=0}^{k-1} |K_{t,j} - K_{m,j}| - \varepsilon_k,$$

with $\varepsilon_k = \mathcal{O}(h^2)$. Thus discrete LEDR cannot converge to zero in the presence of persistent curvature mismatch.

4.8 Practical computation from measured data

Equation (3) suggests a practical estimator for the curvature discrepancy:

$$K_k \approx \frac{2\xi_k - \xi_{k+1} - \xi_{k-1}}{h^2 \xi_k},$$

valid whenever $\xi_k \neq 0$ and $\|\xi_k\|$ is sufficiently small. This estimator can be used directly on flight data or neural network prediction residuals to recover curvature signatures in real time.

5 Curvature-Induced Resonance

This section analyzes the intrinsic oscillatory behavior that arises in the latent error dynamics when the true manifold exhibits positive sectional curvature relative to the model. We show that resonance is not a numerical artifact but a geometric consequence of the Jacobi operator. Both continuous and discrete formulations are considered.

5.1 Continuous-time Jacobi dynamics

From Section 3, the latent error $\xi(t)$ satisfies the continuous-time LEDR equation

$$\frac{D^2 \xi}{dt^2} + R^t(T_m, \xi)T_m = F_{\Delta\Gamma}(T_m, \xi), \quad (5)$$

where the forcing term $F_{\Delta\Gamma}$ arises from connection mismatch.

We first consider the flat-model case ($\Gamma^m \equiv 0$), for which (5) reduces to

$$\frac{D^2 \xi}{dt^2} + R^t(T_m, \xi)T_m = 0. \quad (6)$$

Let $K(t)$ denote the sectional curvature of the plane spanned by $\{T_m(t), \xi(t)\}$. Then the Jacobi operator admits the scalar representation

$$R^t(T_m, \xi)T_m = K(t) \xi(t),$$

whenever ξ is orthogonal to T_m .

Equation (6) becomes

$$\ddot{\xi}(t) + K(t) \xi(t) = 0. \quad (7)$$

Positive curvature implies local geodesic convergence, yielding bounded oscillatory deviations. Negative curvature implies exponential divergence. Zero curvature yields linear drift.

Theorem 5 (Intrinsic MER under positive curvature). *Suppose the sectional curvature satisfies*

$$K(t) \geq K_0 > 0 \quad \forall t.$$

Then all solutions of (7) are oscillatory and satisfy

$$\xi(t) = A(t) \sin\left(\int_0^t \sqrt{K(\tau)} d\tau\right) + B(t) \cos\left(\int_0^t \sqrt{K(\tau)} d\tau\right),$$

where $A(t), B(t)$ vary slowly with $\|\dot{K}\|$. Thus positive curvature necessarily induces persistent oscillatory latent error dynamics (MER).

This establishes MER as a direct geometric effect of curvature.

5.2 Discrete-time MER

The discrete LEDR dynamics from Section 4 take the form

$$\xi_{k+1} = 2\xi_k - \xi_{k-1} - h^2 K_k \xi_k + \mathcal{O}(h^3, \|\xi_k\|^2), \quad (8)$$

under flat-model assumptions.

Neglecting higher-order terms yields

$$\xi_{k+1} - (2 - h^2 K_k) \xi_k + \xi_{k-1} = 0. \quad (9)$$

If $K_k = K > 0$ is constant, the characteristic polynomial is

$$\mu^2 - (2 - h^2 K) \mu + 1 = 0.$$

Its roots satisfy

$$|\mu_{\pm}| = 1 \iff 0 < h^2 K < 4.$$

Thus discrete MER is guaranteed for all sufficiently small sampling steps.

Theorem 6 (MER persists under sampling). *Let $K_k \geq K_0 > 0$ and h sufficiently small ($h < 2/\sqrt{K_0}$). Then every nontrivial solution of (9) is oscillatory with*

$$\xi_k = C_1 \cos(\omega_d k) + C_2 \sin(\omega_d k),$$

where the discrete frequency ω_d is given by

$$\omega_d = \arccos\left(1 - \frac{h^2 K}{2}\right).$$

Hence curvature-induced resonance persists in discrete-time systems.

5.3 Interpretation

The results in this section show that MER is an intrinsic manifestation of curvature mismatch between the true and model connections:

- Positive curvature causes geodesics to bend toward each other.
- A flat model fails to reproduce this bending.
- The resulting geodesic deviation follows the Jacobi structure $\ddot{\xi} + K\xi = 0$.
- The latent error must oscillate, regardless of sampling or numerical integration.

Thus MER is not an artifact of noise, discretization, or specific system structure. It is the natural dynamical signature of curvature on the true manifold.

6 Sphere–Plane Example: Closed-Form LEDR and MER Dynamics

To illustrate the geometric mechanism underlying LEDR and MER, we present a closed-form analysis in the simplest nontrivial setting: a true system evolving on a sphere of constant curvature, while the model assumes a flat Euclidean plane. This setting admits analytical geodesics, curvature tensors, and exact LEDR solutions, enabling a transparent demonstration of curvature-induced resonance and divergence.

6.1 Problem setup

Let the true dynamics evolve on the 2-sphere $\mathbb{S}^2(r)$ of radius r embedded in $\mathbb{R}^3$, with constant sectional curvature

$$K = \frac{1}{r^2}.$$

Let the model assume a flat plane $\mathbb{R}^2$ with Euclidean metric and zero Christoffel symbols,

$$g_m = I_2, \quad \Gamma^m \equiv 0.$$

Let $x_t(t)$ be a true geodesic on $\mathbb{S}^2(r)$ and $x_m(t)$ the “shadow” geodesic generated by the model from the same initial condition. The LEDR displacement is

$$\xi(t) = x_t(t) - x_m(t).$$

6.2 True and model geodesics

On $\mathbb{S}^2(r)$, great circles are geodesics. Using arc-length parameterization, the true solution can take the form

$$x_t(t) = r \begin{bmatrix} \cos(t/r) \\ \sin(t/r) \\ 0 \end{bmatrix},$$

for an initial velocity of unit magnitude.

The model assumes straight-line geodesics on $\mathbb{R}^2$, giving

$$x_m(t) = \begin{bmatrix} r \\ t \end{bmatrix}$$

under the same initial position and velocity projected to the tangent plane at $(r, 0, 0)$.

Because curvature is neglected, $x_m(t)$ drifts linearly while the true trajectory remains bounded.

6.3 Curvature tensor on the sphere

For the sphere, the Riemann curvature tensor takes the classical form

$$R^t(u, v)w = K(\langle w, u \rangle v - \langle w, v \rangle u),$$

and for geodesics with velocity $T = \dot{x}_t$ tangent to the sphere, the Jacobi operator reduces to

$$R^t(T, \xi)T = K\xi,$$

for a deviation vector ξ orthogonal to T .

6.4 Continuous LEDR dynamics

From the continuous LEDR equation derived in Section 3:

$$\frac{D^2\xi}{dt^2} + R^t(T, \xi)T = 0,$$

and using the spherical curvature identity above, we obtain

$$\ddot{\xi}(t) + K\xi(t) = 0. \quad (10)$$

This is a simple harmonic oscillator. Thus curvature mismatch between the sphere and the plane necessarily induces oscillatory LEDR behavior.

6.5 Closed-form LEDR solution

Solving (10) yields

$$\xi(t) = A \sin(\sqrt{K} t) + B \cos(\sqrt{K} t),$$

where A, B are determined by initial LEDR displacement and velocity.

The LEDR frequency encodes curvature:

$$\omega_{\text{LEDR}} = \sqrt{K} = \frac{1}{r}.$$

Thus the LEDR oscillation directly reveals the curvature mismatch between the true dynamics and the model.

6.6 Discrete LEDR dynamics

Applying the discrete LEDR equation derived in Section 4, the recurrence becomes

$$\xi_{k+1} - (2 - h^2 K)\xi_k + \xi_{k-1} = 0.$$

Its characteristic roots satisfy

$$\mu^2 - (2 - h^2 K)\mu + 1 = 0.$$

When h is sufficiently small and $K > 0$ (as on a sphere), the roots are complex conjugates of unit modulus, resulting in persistent oscillation:

$$|\mu_{\pm}| = 1, \quad \arg(\mu_{\pm}) = \arccos\left(1 - \frac{h^2 K}{2}\right).$$

Thus LEDR remains oscillatory under sampling, and MER survives discretization.

6.7 Geometric interpretation

The plane model attempts to integrate a geodesic with zero curvature, while the true manifold imposes curvature $K = 1/r^2$. The two geodesics therefore separate as predicted by the classic Jacobi equation: the plane geodesic drifts, whereas the sphere geodesic bends. LEDR measures this bending discrepancy.

The resulting harmonic LEDR signal represents the curvature-induced deviation of the true geodesic from the flat one. This gives a direct geometric meaning to MER:

MER corresponds to the oscillatory Jacobi response caused by positive sectional curvature on the true manifold.

6.8 Implications

The sphere–plane example demonstrates three key facts:

1) **Curvature mismatch produces an intrinsic dynamic signature.** LEDR necessarily oscillates for $K > 0$.

2) **MER is a geometric phenomenon.** It reflects the curvature-induced acceleration of geodesic separation.

3) **LEDR frequency reveals curvature.** The frequency $\sqrt{K}$ is a direct observable from data.

This example serves as a canonical analytic case supporting the general theory.

7 Discussion

The results presented in this work establish a geometric framework for understanding model error as a dynamical quantity rather than a static residual. By interpreting the model and true systems as evolving on separate affine manifolds with different curvature properties, the LEDR dynamics naturally emerge as a discrete or continuous geodesic deviation equation. This formulation provides several conceptual advantages.

First, curvature mismatch between the model and the true manifold appears explicitly as a forcing term in the Jacobi equation. This reveals that model inconsistency is intrinsically encoded in the second-order error dynamics, and cannot be removed through filtering, parameter tuning, or higher sampling rates. In particular, MER arises as a geometric inevitability when the true manifold exhibits positive sectional curvature along the trajectory.

Second, the LEDR signal contains significantly richer information than traditional first-order or scalar residuals. The oscillatory structure, frequency content, and growth rates of the LEDR trajectory directly reflect the curvature difference between the two connections. This implies that LEDR may serve as a foundation for designing new model-validity tests, system identification procedures, and diagnostic algorithms in a broad class of dynamical systems.

Crucially, the structure of the LEDR equation (1) suggests a spectral classification of model error that goes beyond simple magnitude analysis. The error dynamics consist of a homogeneous part driven by the true curvature (the Jacobi operator) and a particular part driven by the connection mismatch ($F_{\Delta\Gamma}$). This distinction allows us to diagnose the nature of the model deficiency.

When the model is structurally simple (e.g., flat, $\nabla^m \approx 0$), the forcing term $F_{\Delta\Gamma}$ is negligible compared to the curvature restoring force. In this case, the LEDR exhibits pure, harmonic-like oscillations at the natural frequency $\sqrt{K}$ (Intrinsic MER). This "clean" signature indicates that the model is consistent but lacks the necessary geometric curvature terms.

When the model includes complex but incorrect geometric terms ($\nabla^m \neq 0$ and $\nabla^m \not\approx \nabla^t$), the connection mismatch term $F_{\Delta\Gamma}$ acts as a significant external driver. This results in a forced response, potentially leading to beating phenomena or irregular waveforms caused by the interference between the true manifold's natural frequency and the model's incorrect geometric forcing.

Consequently, a clean oscillatory error signal suggests one should add curvature to the model, whereas a complex, beating error signal suggests one must correct the underlying geometric structure.

Third, the framework unifies a variety of phenomena traditionally viewed as unrelated. Oscillatory model error in sampled systems, persistent residual dynamics in model-based predictors, and certain forms of optimization instability can all be interpreted as manifestations of geodesic deviation under curvature mismatch. This suggests a wide applicability of the LEDR formalism beyond the examples presented here.

Finally, while the analysis in this paper focuses on local Jacobi dynamics, the framework naturally extends to higher-order geometric effects, including connection mismatch, curvature gradients, and nonlinear manifold deformation. These directions remain open for future investigation.

Perhaps the most significant implication of this framework is the potential to shift the paradigm of predictive

modeling. Currently, the pursuit of high-fidelity models often involves adding complexity (e.g., more parameters, finer meshes, deeper neural networks) to minimize static residuals. Our theory suggests an alternative: one could start with a structurally simple (e.g., flat or constant-curvature) model and then actively compensate for model error by *estimating and correcting for the curvature mismatch in real time*. The LEDR signal, as shown in our discrete formulation, contains the necessary information to estimate the local sectional curvature of the true manifold. This opens the door to a new class of "curvature-informed" adaptive predictors that do not just react to errors, but actively correct the underlying geometry of the model dynamics. Such a system would be inherently more robust and efficient, requiring only a coarse initial model augmented by a dynamic geometric compensator.

This geometric perspective is particularly promising for infinite-dimensional systems, such as fluid dynamics governed by the Navier-Stokes equations. Since ideal fluid flow can be viewed as geodesic motion on the group of diffeomorphisms [3], the turbulence modeling problem—typically treated by adding empirical viscosity terms—could be reimaged as a problem of compensating for the sectional curvature mismatch between the resolved scales (model) and the unresolved scales (truth). Our LEDR framework offers a principled way to quantify and correct this geometric discrepancy.

8 Conclusion

This paper develops a geometric theory of model error dynamics based on affine connections and geodesic deviation. By treating the model and true systems as evolving on distinct manifolds with potentially different curvature, we show that the latent error dynamics satisfy a Jacobi-type equation in both continuous and discrete time. This provides an explicit link between curvature mismatch and the emergence of oscillatory or divergent model-error behavior.

A key consequence of this formulation is that MER arises as an intrinsic response of the Jacobi operator whenever the true manifold exhibits positive sectional curvature relative to the model. Thus MER is not merely a numerical artifact or an unstable mode, but a fundamental geometric signature of model-inconsistency. Moreover, the LEDR signal furnishes a direct observable that encodes curvature information and can be computed without explicit knowledge of the true system.

The sphere-plane example demonstrates that the theory admits closed-form solutions and yields immediately interpretable curvature-dependent behavior. The discrete-time derivation further confirms that the geometric structure is preserved under sampling and can

be exploited in digital controllers, system identification, and model-based prediction frameworks.

Overall, the results establish LEDR and MER as fundamental tools for analyzing model fidelity. They enable a shift from static, residual-based diagnostics to a dynamic, curvature-informed consistency analysis. The ultimate promise of this approach lies in the creation of a new generation of predictive systems: instead of pursuing ever-increasing model complexity, one could employ a coarse baseline model and dynamically compensate for its structural deficiencies by estimating and correcting for curvature mismatch from the observed LEDR signal. Future work will focus on developing robust algorithms for real-time curvature reconstruction and exploring the applications of this geometric compensation paradigm to control, estimation, and learning-based dynamical models.

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A Derivation of the Continuous LEDR Equation

Consider a smooth manifold $\mathcal{M}$ endowed with two affine connections: the true connection ∇^t and the model connection ∇^m . Let $x_t(t)$ and $x_m(t)$ denote the true and model trajectories satisfying

$$\frac{DT_t}{dt} = \nabla_{T_t}^t T_t, \quad \frac{DT_m}{dt} = \nabla_{T_m}^m T_m,$$

where $T_t = \dot{x}_t$ and $T_m = \dot{x}_m$.

Define the LEDR displacement field

$$\xi(t) = x_t(t) - x_m(t).$$

We linearize the true geodesic equation around the model trajectory. Let $x_\epsilon(t)$ be a geodesic variation

$$x_\epsilon(t) = x_m(t) + \epsilon \xi(t) + \mathcal{O}(\epsilon^2),$$

with velocity

$$T_\epsilon(t) = T_m(t) + \epsilon \dot{\xi}(t) + \mathcal{O}(\epsilon^2).$$

Substituting into the true geodesic equation

$$\nabla_{T_\epsilon}^t T_\epsilon = 0$$

and expanding to first order in ϵ yields:

$$\nabla_{T_m}^t \dot{\xi} + \nabla_{\xi}^t T_m + R^t(T_m, \xi)T_m = 0,$$

where we used the curvature identity

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = R(X, Y)Z.$$

Using the symmetry $\nabla_{\xi}^t T_m = \nabla_{T_m}^t \dot{\xi}$ for a torsion-free connection, we obtain the classical Jacobi operator:

$$\frac{D^2 \xi}{dt^2} + R^t(T_m, \xi)T_m = 0,$$

where $\frac{D}{dt}$ denotes the true covariant derivative along $x_m(t)$.

Finally, because the model connection differs from the true one by

$$\Delta \Gamma = \Gamma^t - \Gamma^m,$$

additional forcing terms arise when expressing $\frac{D}{dt}$ using ∇^m . Collecting all first-order terms gives the LEDR equation:

$$\frac{D^2\xi}{dt^2} + R^t(T_m, \xi)T_m = F_{\Delta\Gamma}(T_m, \xi).$$

This completes the derivation.

B Derivation of the Discrete LEDR Equation

Let $x_{t,k}$ and $x_{m,k}$ denote the true and model states sampled at $t_k = kh$ with fixed step size h . Define the discrete LEDR displacement

$$\xi_k = x_{t,k} - x_{m,k}.$$

B.1 Discrete covariant derivatives

We use second-order consistent central difference approximations:

$$\left. \frac{D\xi}{dt} \right|_{t_k} = \frac{1}{2h}(\xi_{k+1} - \xi_{k-1}) + \mathcal{O}(h^2),$$

$$\left. \frac{D^2\xi}{dt^2} \right|_{t_k} = \frac{1}{h^2}(\xi_{k+1} - 2\xi_k + \xi_{k-1}) + \mathcal{O}(h^2).$$

B.2 Discretization of the curvature term

The continuous curvature action

$$R^t(T, \xi)T$$

is discretized by evaluating the curvature tensor at $x_{m,k}$:

$$[R_k^t(T_{m,k}, \xi_k)T_{m,k}]^i = R^{t\ i}_{j\ell m}(x_{m,k})T_{m,k}^j \xi_k^\ell T_{m,k}^m,$$

where the discrete velocity is [12]

$$T_{m,k} = \frac{x_{m,k+1} - x_{m,k-1}}{2h}.$$

B.3 Discretization of the LEDR equation

Substituting the discrete derivatives into the continuous LEDR equation

$$\frac{D^2\xi}{dt^2} + R^t(T_m, \xi)T_m = F_{\Delta\Gamma}(T_m, \xi),$$

we obtain

$$\frac{1}{h^2}(\xi_{k+1} - 2\xi_k + \xi_{k-1}) + R_k^t(T_{m,k}, \xi_k)T_{m,k} = F_{\Delta\Gamma,k}(T_{m,k}, \xi_k) + \mathcal{O}(h^2).$$

Multiplying by h^2 and rearranging yields the discrete LEDR evolution:

$$\begin{aligned} \xi_{k+1} = & 2\xi_k - \xi_{k-1} - h^2 R_k^t(T_{m,k}, \xi_k)T_{m,k} \\ & + h^2 F_{\Delta\Gamma,k}(T_{m,k}, \xi_k) + \mathcal{O}(h^3, \|\xi_k\|^2). \end{aligned}$$

B.4 Constant curvature case

When the true manifold has constant sectional curvature K ,

$$R^t(T, \xi)T = K\xi,$$

and the recurrence reduces to

$$\xi_{k+1} - (2 - h^2 K)\xi_k + \xi_{k-1} = 0.$$

This is the standard second-order central finite-difference approximation of the Jacobi equation

$$\ddot{\xi} + K\xi = 0.$$

B.5 Discrete stability and MER

The characteristic equation

$$\mu^2 - (2 - h^2 K)\mu + 1 = 0$$

has complex roots when

$$0 < h^2 K < 4,$$

which yields oscillatory behavior (MER). For $K > 0$, oscillations are guaranteed for sufficiently small h .

This proves that discrete LEDR reproduces MER in sampled systems.

C Proof of the Curvature Obstruction Theorem (Theorem 3)

Theorem 3 states that a persistent mismatch in sectional curvature prevents the latent error $\xi(t)$ from uniformly converging to zero. We provide a sketch of the proof based on comparison principles for second-order differential equations.

Consider the general LEDR equation from Theorem 4:

$$\frac{D^2\xi}{dt^2} + R^m(T^m, \xi) T^m = -\Delta R(T^m, \xi) T^m + F_{\Delta\Gamma}(T^m, \xi).$$

Let $K_t(t)$ and $K_m(t)$ be the sectional curvatures of the true and model connections along the plane spanned by T^m and ξ . The equation can be written in scalar form for the magnitude $\|\xi(t)\|$ as a differential inequality:

$$\|\ddot{\xi}\| + K_m(t)\|\xi\| \approx -(K_t(t) - K_m(t))\|\xi\| + \text{h.o.t.}$$

Let $\Delta K(t) = K_t(t) - K_m(t)$. The equation for the error dynamics is approximately

$$\|\ddot{\xi}\| + K_t(t)\|\xi\| \approx 0.$$

If we consider the difference between the true dynamics and the model dynamics acting on ξ , the core driving term is proportional to $\Delta K(t)$.

By the Sturm-Picone comparison theorem, if $K_t(t) > K_m(t)$, the solutions to the true system's Jacobi equation oscillate faster than those of the model system. This mismatch in oscillatory frequency prevents the error from being nulled.

More formally, integrating the differential inequality twice, and applying a Gronwall-type argument, shows that $\|\xi(t)\|$ is driven by the integral of the curvature mismatch. If $|\Delta K(t)| \geq \kappa_0 > 0$, then

$$\|\xi(t)\| \geq C \int_0^t \int_0^s |\Delta K(\tau)| \|\xi(\tau)\| d\tau ds - \varepsilon(t),$$

which implies that $\|\xi(t)\|$ cannot decay to zero if it is initially nonzero. The integral of the curvature mismatch acts as a persistent energy source for the error dynamics, establishing the lower bound stated in the theorem. A full treatment requires careful handling of the covariant derivatives and projection operators, but this captures the essence of the obstruction.